

Seasonal and spatial dynamics of the aquatic insect communities of an intermittent Mediterranean river

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ABSTRACT

Although intermittent rivers are present in all climate regions, their biodiversity and ecosystem functions were largely overlooked in the past. Recently, considerable effort has been made to improve our understanding of these, however, many data gaps still remain. We conducted a study to improve knowledge of the community patterns and processes in intermittent rivers in the Dinaric Western Balkan ecoregion. We studied spatial and temporal variation of aquatic insect community composition and structure in the intermittent Krčić River over two seasons (June and December) in three river sections (spring, upper/middle and lower reaches), encompassing dominant microhabitats (phytal, boulders, cobbles, sand and xylal). Community was primarily composed of aquatic insects (92 % of the community) and seasonality was inferred as the most important factor determining its composition, diversity and abundance. This was also supported by the ecological trait information, i.e. trophic and dispersal-related community structure. The most diverse and abundant insect group were the under-investigated chironomids, thus, we discussed tolerance and intermittency indicator potential of some of these taxa. By providing new insights into temporal and spatial dynamics of aquatic insect communities, a major bioindicator component of the macroinvertebrate community used in water quality monitoring in intermittent rivers, our data support development of the intermittent river-specific and adequate management and monitoring procedures.

1. Introduction

Intermittent rivers (IR) are highly dynamic and complex ecosystems, characterized by flow intermittence. During dry seasons, the flow progressively ceases and the riverbed becomes a terrestrial habitat which can contain isolated pools (Stubbington et al., 2017). Gradual loss of water in IR changes the structure and composition of biotic communities. Thus, during the flowing period lotic generalist and colonizing species dominate in the community (Bogan et al., 2015), but the progressive drying causes a decrease in the number of lotic species and an increase in lentic species. During the dry phase, a community is composed of colonizing terrestrial species, and a few lentic and lotic species surviving the unfavorable period as resistant eggs, as larvae or adults buried in the sediment and as terrestrial adults (Stubbington et al., 2017). Furthermore, the recolonization process after rewetting is

an important mechanism for maintaining metapopulation and meta-community dynamics (Datry et al., 2017a). It is largely dependent on dispersal potential that is affected by multiple traits relating to the morphology, life history and behavior of different life stages (Sarremejane et al., 2020).

Communities in IR are usually subsets of the communities from adjacent perennial rivers and streams with the domination of generalist and colonizing species (Corti and Datry, 2016; Stubbington et al., 2017). These communities are vulnerable since the isolation of populations leads to adaptation and specialization of certain species, which is why endemism can be increased (Stubbington et al., 2017).

IR have a great influence on the hydrology of adjacent perennial water bodies and are heavily affected by climate change, but poor understanding of their function and role in ecosystems results in the high rate of their degradation (Acuña et al., 2014; Datry et al., 2017b;

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Stubbington et al., 2018). In recent years international actions have been launched to develop a research network for gathering knowledge and improving the understanding of these systems (Datry et al., 2017b; Leigh et al., 2017; Prat et al., 2018). In spite of the considerable improvement, there is still a large gap in the knowledge on response of IR communities to environmental impacts, such as drying (e.g. Straka et al., 2019; Stubbington et al., 2019). Better knowledge of IR community responses is important for predicting future events and to develop adequate and effective management strategies and policies (Corti and Datry, 2016). More precisely, information on biological species traits (e.g. capacity to use drought refuges, dispersal traits) is vital when attempting to predict how stream communities and their populations will respond to increased drying due to climate change and/or other anthropological impacts (Robson et al., 2011).

In the European Union, there is no unified management system for IR. Since many IR are not recognized as “water bodies” by the EU Water Framework Directive, 2000/60/EC (EC 2000) due to their small size, implementation at the EU level is not defined (Acuña et al., 2014). Also, biomonitoring is based only on the aquatic part of the component excluding terrestrial community (EC 2000). Even so, some abundant aquatic taxa groups (e.g. the Chironomidae, Diptera) are used with a relatively coarse taxonomic resolution (Birk et al., 2012; Schmidt-Kloiber and Nijboer, 2004). Therefore, there is a great need for the development of a detailed pressure-impact relationship model due to various responses to a range of human pressures (Magand et al., 2020).

The main aim of the current study was to improve understanding of ecological patterns and processes in IR in the European Mediterranean regions, particularly the under-investigated Dinaric Western Balkan ecoregion (Illies, 1967), by looking into spatial and temporal dynamics of aquatic insect communities. Our first objective was to investigate the variation in composition and structure of the communities among river sections, seasons and microhabitat types. We expect the communities to

show greatest differences between seasons due to life cycles adapted to flow intermittence and seasonal variation in food availability. With the emphasis on Chironomidae, as the most abundant but insufficiently studied group, we provide novel insights into dynamics of their community composition and structure in IR. Our second objective was to assess the effects of major environmental drivers on communities of dominant aquatic insect groups (Ephemeroptera, Plecoptera, Trichoptera, Diptera: Chironomidae). Distance from the spring and water temperature are expected to have the highest impact on aquatic insect distribution. Finally, we use ecological trait information to analyse spatial and temporal differences in trophic and multiple dispersal-related community structure of aquatic insects. We expect both to vary mainly in regard to the season, depending on the prevalence of different food types and differences in life cycles and dispersal strategy of aquatic insects, respectively. By providing new insights into temporal and spatial dynamics of aquatic insect communities in intermittent rivers, our data support development of the IR-specific and adequate monitoring procedures.

2. Materials and methods

2.1. Study area

This research was conducted along the intermittent karst river Krčić, situated in Mediterranean part of Croatia, in the Dinaric Western Balkan ecoregion (Fig. 1). Krčić is about 10 km long, and its course usually dries up during the summer and early autumn months. Dry period varies from September to December, depending on the arrival of first heavy autumn/winter rains, and duration of dry phase differs along the course (Bonacci et al., 2006). According to observations in 1980's, residual pools remained longest in the lower reaches, and in most years the whole course dried up for a certain period (Bonacci, 1985). Our recent

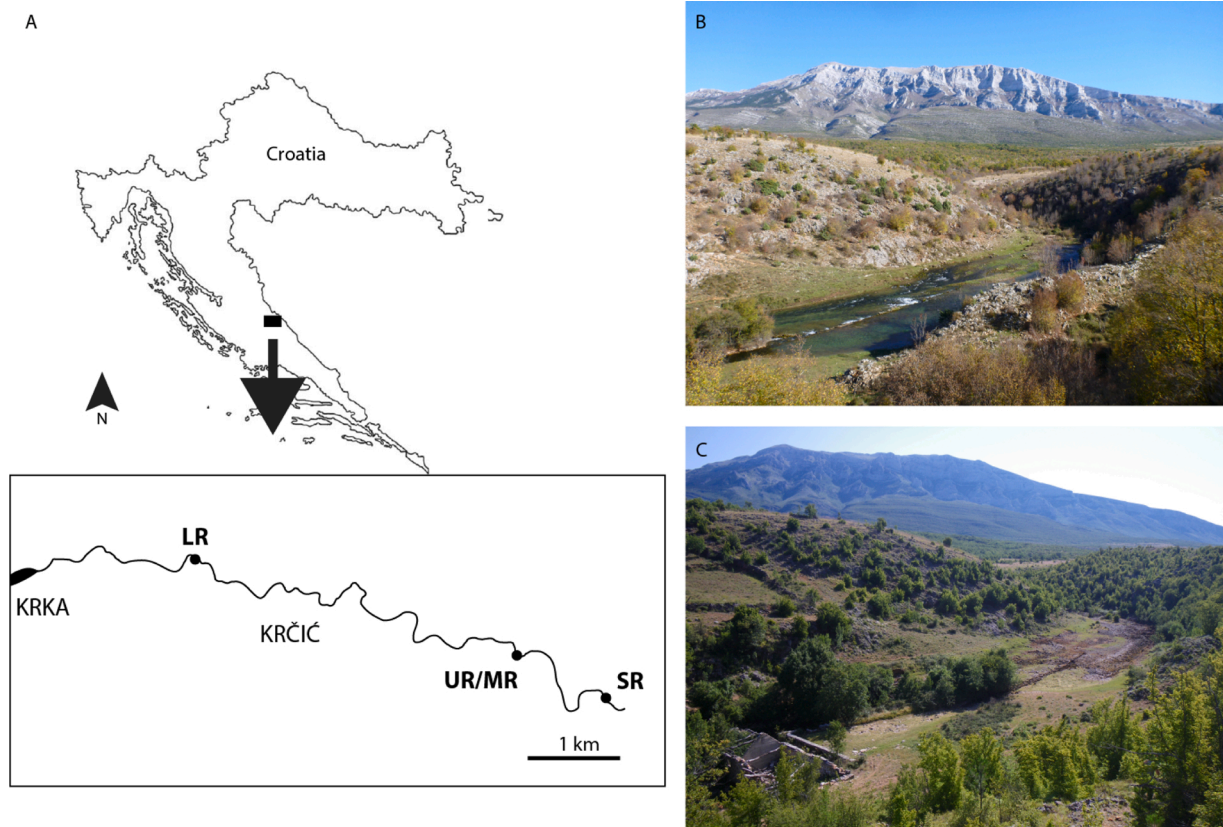


Fig. 1. A) Map of the sampling sites at the Krčić River (SR – spring reaches, UR/MR – upper reaches/middle reaches, LR – lower reaches) and B) flowing phase and C) dry riverbed of the spring reaches of the Krčić River.

observations (from 2014 onwards, A.B. unpublished data) suggest that residual pools remain longest in the upper and middle reaches, whereas lower reaches dry out faster. However, no gauges are present, thus the obviously complex patterns cannot be verified with precise measurements of discharge patterns. Riverbed and the catchment area are situated on limestone and dolomite bedrock, which results in strong karstification of the riverbed and the great loss of surface water. There are no major hydromorphological alterations, i.e. no agriculture and water abstraction for irrigation, wastewater effluent or settlements in the catchment area. Therefore, Krčić River is minimally influenced by anthropogenic activities and included in the Natura 2000 network. Thus, it has a potential for being regarded a reference site for IRs in this region.

2.2. Field sampling and sample processing

Benthic macroinvertebrate community sampling was conducted in the Krčić River in two seasons during the flowing phase, in June and December 2016. Sampling was conducted approximately 230 days and 45 days after the last dry phase, in June and December, respectively. River flow is divided into three main parts (Fig. 1, Table 1): spring reaches (SR) – characterized by the presence of macroinvertebrates typically occurring in springs (such as *Synurella ambulans* Müller; (Zutinić et al., 2018) upper/middle reaches (UR/MR) – representing the dominant part of river morphology, characterized by stony riverbed covered with moss and open-canopy riparian vegetation; lower reaches (LR) – characterized by sandy riverbed and coarse organic matter - xylal as a consequence of closed-canopy riparian vegetation (poplar forest). In each section, three dominant microhabitats were identified (SR: moss, macrophytes, boulder/cobbles; UR/MR: moss, sand, boulder/cobbles; LR: sand, xylal, moss and boulders). Macroinvertebrate samples were taken using a Surber sampler (frame dimensions 25 × 25 cm; mesh size 500 µm) in triplicates per each selected microhabitat per site, in two seasons, thus, a total of 54 samples were collected (Table S1). Additionally, the basic physico-chemical parameters were measured: water temperature, dissolved oxygen, oxygen saturation (WTW Oxi 330 probe); pH (WTW pH 341i probe); conductivity (WTW Cond 340i probe) and alkalinity (measured by titration of the sample with hydrochloric acid (HCl) using methylene chloride as an indicator) (Table 1).

In the laboratory, macroinvertebrates were sorted according to taxonomic groups. The most abundant taxonomic groups contributing to the total community with more than 1.5 % (Chironomidae – Diptera, Ephemeroptera, Plecoptera and Trichoptera; EPT, Table S2) were identified to the lowest taxonomic category possible, using standard identification keys (Bauernfeind and Humpesch, 2001; Jacob, 1974; Moller Pillot, 1984; Waringer and Graf, 2011; Zwick, 2004).

2.3. Data analysis

Abundance was expressed as the number of individuals per m². The

following indices were calculated from the recorded macroinvertebrate community listed in Table S2 using Asterics 4.0.4 (AQEM Consortium, 2002): Shannon's and Simpson's diversity indices, Pielou's evenness index, EPT and EP indices. Generalized linear models (GZLMs) were constructed in SPSS Statistics 27 (IBM Corp., 2020) to test the influence of microhabitats, river sections and seasons on analyzed indices, number of taxa and abundance. A separate model with three independent variables (season, river section and microhabitat) was created for each community parameter. Shannon diversity index and the number of taxa (after logarithmic transformation) achieved normal distribution and linear scale response was used. For other parameters normal distribution was not achieved despite transformations, so gamma scale response with log link was used. Maximum likelihood estimate was used for parameter estimation. Pairwise contrasts of estimated means were performed using Wald's statistics.

Similarity between the overall macroinvertebrate community in two sampling seasons and three different river sections was investigated using nonmetric multidimensional scaling analysis (nMDS) based on Bray-Curtis similarity index with superimposed hierarchical clustering analysis (using a complete linkage option) in Primer 7.0 (Clarke and Gorley, 2015).

The program Asterics 4.0.4 (AQEM Consortium, 2002) was used to calculate trophic structure of the macroinvertebrate community, i.e. the functional feeding guilds (Moog and Hartmann, 2002). Another set of GZLMs were constructed to test for differences between proportion of functional feeding guilds within macroinvertebrate community at three different levels (seasons, river sections and combination season-river sections, as independent variables). Only the proportion of gatherers/collectors (not transformed) achieved normal distribution and linear scale response was used, while for other parameters gamma scale response with log link was used. Maximum likelihood estimate was used for parameter estimation. Pairwise contrasts of estimated means were performed using Wald's statistics.

The relationship between environmental factors and aquatic insect community composition was assessed using the Canonical Correspondence Analysis (CCA) in the CANOCO version 5.11 package (ter Braak and Smilauer, 2012). The analysis was conducted separately for Chironomidae and EPT, as these were the dominant aquatic insect groups (Table S2) in order to enable more detailed, group specific focus of inferences. All EPT and chironomid taxa were included in the analysis (taxa and corresponding abbreviations listed in Table S1). Prior to the CCA analyses, the abundances of EPT and chironomids were log transformed and rare taxa were down-weighted due to occurrence of rare taxa with a low total abundance. In the CCA, to identify the most important environmental predictors that significantly explained variance in the EPT and chironomid data, redundant variables were removed by a stepwise forward selection procedure. The significance of the correlations between taxa occurrence and environmental variables was tested using the Monte Carlo test, with 999 permutations ($p < 0.05$).

Quantitative representation of dispersal related community structure

Table 1

Geographic position of sampling sites and physico-chemical properties of water in the Krčić River in two seasons in 2016. SR – spring reaches, UR/MR – upper/middle-reaches, LR – lower reaches.

Geographic position	SR		UR/MR		LR	
Geographic coordinates	N44.026787	E16.333054	N44.029260	E16.309626	N44.043218	E16.254543
Altitude		406 m		354 m		282 m
Distance from spring (m)		350.00		2280.00		8630.00
Physico-chemical properties	June	December	June	December	June	December
Water temperature (°C)	9.37	8.97	10.10	8.93	11.20	8.83
Dissolved oxygen (mg/L)	11.31	11.56	11.62	11.05	10.14	11.53
Oxygen saturation (%)	106.83	102.30	107.70	97.87	97.43	101.47
Conductivity (µS/cm)	391.33	352.00	380.00	345.33	350.33	339.00
pH	7.52	7.79	8.25	8.35	8.40	8.49
Alkalinity	4.00	4.20	4.10	4.00	4.00	4.17
Depth (cm)	19.89	30.00	22.89	23.67	18.78	22.78

was obtained using data on nine dispersal-related biological traits (maximum body size, life cycle duration, potential number of reproductive cycles per year, dispersal strategy, adult life span, female wing length, wing pair type, lifelong fecundity, propensity to drift) available in the DISPERSE database at the genus and family level for aquatic insect taxa (Sarremejane et al., 2020). As with the latter analysis, it was also conducted for the most abundant insect groups, i.e. Chironomidae and the EPT orders. Proportion of each trait category in the sample was calculated following the procedure in (Moog and Hartmann, 2002), briefly, it represents the sum of all products (trait category $\times$ proportion) for all present taxa, for total proportion of taxa with available dispersal-data in the sample. Differences in proportions among seasons, river sections and microhabitats were then tested using GZLMs in SPSS Statistics 27 (IBM Corp., 2020). Separate models were constructed for each dispersal-related biological trait using linear or gamma scale response, depending on normality of the data. Results were visualized using the Principal correspondence analysis (PCA) using Primer ver7 (Clarke and Gorley, 2015).

3. Results

Aquatic insect groups accounted for 92 % of total abundance of macroinvertebrates in the Krčić River (Table S2). The most abundant and diverse group of aquatic insects in the Krčić River was the Chironomidae family (Diptera) with a proportion of 69 % and 27 taxa, and with *Microsetra* sp. as the dominant taxon (Table S2). Other insect groups were markedly less represented, containing only a few taxa each (Ephemeroptera – proportion of 7.4 % and three species; Plecoptera – proportion of 9.7 % and four genera, Trichoptera – proportion of 1.8 % and three species; Table S2). The most abundant and most common Trichoptera and Ephemeroptera species were *Micropterna nycterobia* (McLachlan, 1875) – proportion of 1.7 % and *Baetis rhodani* (Pictet, 1843) – proportion of 7.3 %, respectively. Remaining insect groups (orders/dipteran families) accounted for less than 1 % of the community. Apart from aquatic insects, Amphipoda were present in relatively high abundance in the SR (*Synurella* sp.), whereas Oligochaeta were abundant in the LR (Table S2). Aquatic macroinvertebrate abundance and community composition changed along the river course. The highest macroinvertebrate abundance was recorded in UR/MR and the lowest in LR (Table 2). However, taxa richness showed the opposite trend, with the highest value recorded in the LR and the lowest in the SR (Table 2).

3.1. Seasonal and spatial dynamics of aquatic insect community in the Krčić River

According to GZLMs, river section, season and microhabitat all significantly influenced the composition of EPT communities in the Krčić River (i.e. the EPT [number of Ephemeroptera, Plecoptera and Trichoptera] and EP indices [number of Ephemeroptera and Plecoptera], Table 3). Similarly, EP% and EPT% were significantly

Table 3

Main effects of microhabitat, river section and season on the analyzed parameters of aquatic insect community in the Krčić River tested using the Generalized linear models (GZLMs; N - number of taxa; D - Simpson's Index; H - Shannon's Index; J' - Pielou's Index; EPT - number of Ephemeroptera, Plecoptera and Trichoptera taxa; EP - number of Ephemeroptera and Plecoptera taxa; EPT% - proportion of Ephemeroptera, Plecoptera and Trichoptera taxa in community; EP - proportion of Ephemeroptera and Plecoptera taxa in community).

Parameter	Factors	Wald Chi-Square	d.f.	p-value
Abundance	Microhabitat	36.428	5	> 0.001
	River section	1.746	2	0.418
	Season	30.623	1	> 0.001
N	Microhabitat	22.182	5	> 0.001
	River section	2.137	2	0.344
	Season	14.49	1	> 0.001
J'	Microhabitat	1.395	5	0.925
	River section	1.118	2	0.572
	Season	1.205	1	0.272
H	Microhabitat	9.569	5	0.088
	River section	1.803	2	0.406
	Season	0.677	1	0.411
D	Microhabitat	3.291	5	0.655
	River section	1.164	2	0.559
	Season	0.013	1	0.911
EP	Microhabitat	19.118	5	0.002
	River section	19.929	2	> 0.001
	Season	5.735	1	0.017
EP%	Microhabitat	3.471	5	0.628
	River section	12.263	2	0.002
	Season	6.424	1	0.011
EPT	Microhabitat	15.723	5	0.008
	River section	16.522	2	> 0.001
	Season	4.423	1	0.035
EPT%	Microhabitat	3.323	5	0.650
	River section	15.498	2	> 0.001
	Season	19.502	1	> 0.001

influenced by the season and the river section (Table 3). EP% and EPT% was significantly higher in December than in June, while the number of EPT taxa showed the opposite trend (Table S3). SR had significantly lower EP% and EPT% compared to UR/MR and LR (Table S4). Moreover, number of taxa (N) and abundance of macroinvertebrates significantly differed between the two seasons sampled (Table 2), with significantly lower values in December than in June (Table S3). With regard to the microhabitats, macroinvertebrate abundance was significantly higher on moss and sand, and the highest number of EP and EPT taxa were found on boulder and sand, both significantly different to xylal, respectively (Table S5). The number of taxa was lowest on cobbles-boulder and xylal, and highest on sand (Table S5). Diversity indices were not significantly influenced by any tested factor (Table 3).

The nMDS analysis indicated that macroinvertebrate community differed primarily between seasons, at the 40 % similarity level, rather than between different river sections (Fig. 2). Within a given season, only SR clearly stood out with a similarity of 60 %, while UR/MR and LR showed no clear separation between their respective communities.

Table 2

Values of macroinvertebrate community parameters: mean abundance (N of individuals/m²), number of taxa (N) and indices (D - Simpson's Index; H - Shannon's Index; J' - Pielou's Index; EPT - number of Ephemeroptera, Plecoptera and Trichoptera taxa; EP - number of Ephemeroptera and Plecoptera taxa; EPT% - proportion of Ephemeroptera, Plecoptera and Trichoptera taxa in community; EP - proportion of Ephemeroptera and Plecoptera taxa in community) for macroinvertebrate community in the river Krčić according to season (Jun - June; Dec - December) and river section (SR - spring reaches; UR/MR - upper reaches/middle reaches; LR - lower reaches).

Data level	Abundance	N	D	H	J'	EPT%	EP%	EPT	EP
Date									
Jun	25 468	42	0.851	2.473	0.625	12.033	11.943	9	6
Dec	6 047	52	0.911	2.759	0.733	47.933	38.788	11	8
River section									
SR	14 735	40	0.871	2.527	0.685	11.666	10.139	9	6
UR/MR	25 411	45	0.806	2.196	0.577	19.514	17.884	9	7
LR	7 128	47	0.924	2.962	0.769	31.811	28.654	11	8

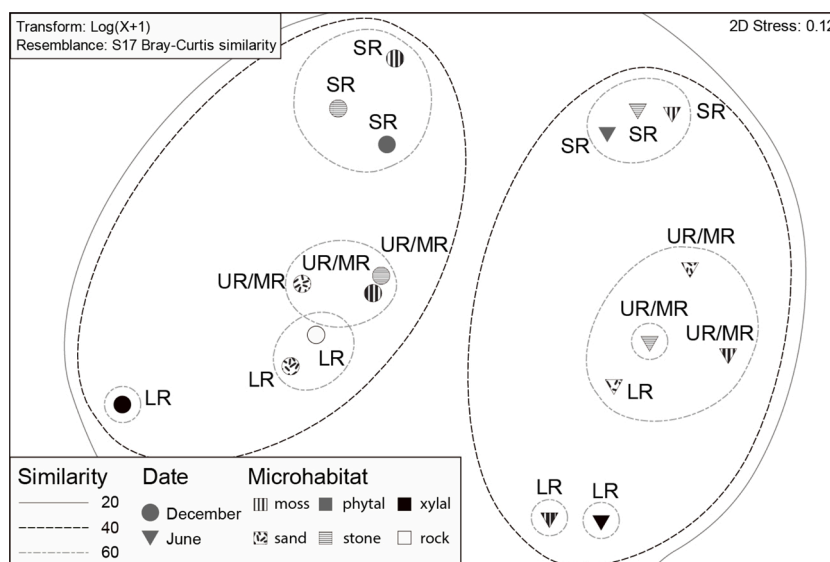


Fig. 2. nMDS analysis of the macroinvertebrate community in two sampling seasons and three different river sections at the Krčić River (SR – spring reaches; UR/MR – upper reaches/middle reaches; LR – lower reaches). Similarity (in %) from hierarchical clustering analysis is superimposed.

Microhabitats did not show an important role in the community grouping.

The CCA analysis shows the relationships between the measured environmental factors and each of the Chironomidae and EPT communities, respectively (Fig. 3 a, b). Permutation test showed that the ordination was statistically significant for both data sets (Chironomidae: $F = 0.5$, $p = 0.003$; EPT: $F = 1.4$, $p = 0.001$). The eigenvalues obtained from the data for the Chironomidae community were 0.298 and 0.260, explaining 60.15 % of the relationship between Chironomidae composition and environmental factors (Fig. 3 a). On the other hand, the eigenvalues obtained from the data for the EPT community were 0.377 and 0.103, explaining 62.30 % of the relationship between EPT taxa composition and environmental factors (Fig. 3 b). In the CCA ordination, water temperature was a good predictor of the composition of both tested communities. However, for Chironomidae the most significant factor was conductivity, while pH value and distance from spring were of the highest importance for EPT (Fig. 3). Community composition of both groups of aquatic insects did not show similarity between river sections, in accordance with the results of the nMDS analysis (Figs. 2 and

3).

3.2. Aquatic insect community trophic structure

The results of the Generalized linear models show highly significant difference in trophic structure of aquatic insects in regard to season, river section and its combination (Table 4). The change of proportion of all functional feeding groups was significant when combination of two factors (season-river section) was used (Table 4). Moreover, shredders, gatherers/collectors and both filter feeders (active and passive), were the functional feeding groups with significant differences between all comparisons (Table 4). Gatherers were the most represented feeding group, consisting mostly of species from the family Chironomidae (especially *Micropsectra* sp., which had the highest population density). They were followed by grazers and scrapers, represented by Ephemeroptera (mostly *B. rhodani*) and Trichoptera (mostly *M. nycterobia*). The proportion of grazers and xylophages remained unchanged across the seasons, while the proportion of predators, represented mostly by Plecoptera (*Isoperla* sp.), did not show significant differences between river

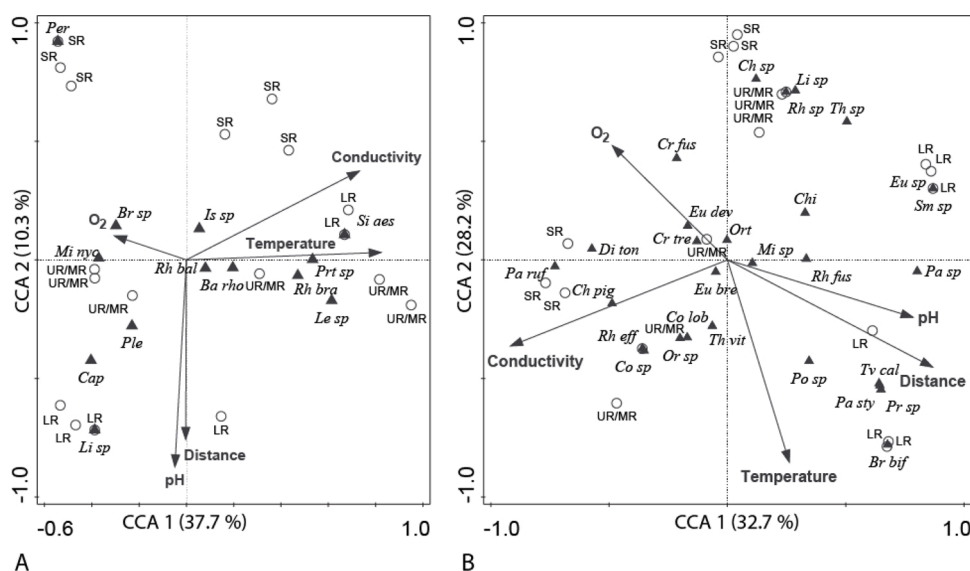


Fig. 3. CCA ordination of environmental variables, river reaches and composition of A) EPT and B) Chironomidae. Taxa are marked with triangles (taxa abbreviations are explained in Table S1), while river sections with circles (SR – spring reaches; UR/MR – upper reaches/middle reaches; LR – lower reaches). Environmental variables are shown with arrows where the directions indicate correlations and the length relative importance of the variables (O_2 – oxygen saturation in %, Distance – distance from the spring in km). Taxa abbreviations are listed in Table S2.

Table 4

Results of the Generalized linear models of proportions of the functional feeding groups at three different levels (two seasons, three river sections and a combination of season and river sections) in the river Krčić (GRA - grazers; XYL - xylophagus taxa; SHR - shredders; GAT/COL - gatherers/collectors; AFF - active filter feeders; PFF - passive filter feeders; PRE - predators; ND - no data available). Significant values are indicated in bold.

Parameter	Independent variables	Wald Chi-Square	d.f.	p-value
GRA	Season-river section	18.216	5	0.003
	River section	14.02	2	0.002
	Season	3.22	1	0.0636
XYL	Season-river section	12.212	5	0.032
	River section	8.323	2	0.011
	Season	4.335	1	0.012
SHR	Season-river section	85.664	5	>0.001
	River section	12	2	0.002
	Season	41.249	1	>0.001
GAT/COL	Season-river section	13.476	5	0.019
	River section	8.584	2	0.049
	Season	1.489	1	0.022
AFF	Season-river section	–	–	–
	River section	1.026	2	0.599
	Season	6.756	1	0.006
PFF	Season-river section	12.103	3	0.007
	River section	14.713	2	0.387
	Season	15.67	1	0.004
PRE	Season-river section	65.575	5	>0.001
	River section	2.689	2	0.261
	Season	11.539	1	0.002
ND	Season-river section	30.414	5	>0.001
	River section	12.527	2	0.002
	Season	5.935	1	0.067

sections (Fig. 4, Table 4).

3.3. Aquatic insect dispersal-related community structure

Aquatic insect community structure based on various dispersal-related biological traits differs primarily in regard to the season, as all nine individual traits significantly differ in proportion in the community between June and December (Generalized linear models; Tables S6, S7). Consequently, the PCA results account for most of the variability between the seasons, with the PC1 component accounting for 44.6 % of variation, separating all samples from Jun and Dec, with exception of one sample (from moss) in the LR in June (Fig. 5). However, proportion of individual trait categories was also inferred to differ significantly between river sections for at least some categories in all traits, except

maximum body size and life cycle duration (Generalized linear models; Tables S6, S7). Hence, in June there is a significantly higher proportion of individuals with larger maximum body size and significantly lower proportion of taxa with life cycle duration ≤ 1 then in December, regardless of the river section (Generalized linear models; Tables S6, S7). Dispersal strategy showed a mixed pattern (Generalized linear models; Tables S6, S7). For the passive dispersal modes (both, aquatic and aerial) seasonal differences were inferred significant, e.g. significantly higher proportion of aerial passive dispersal strategy was observed in December than in June. On the contrary, for active dispersal modes differences between river sections were inferred significant, hence, highest proportion of both active dispersal modes (aquatic and aerial) was observed in the UR/MR and lowest in the SR (Generalized linear models; Tables S6, S7). For the majority of remaining individual traits, season and river section were both inferred to influence differences within particular categories, but general trends are not easily identified.

4. Discussion

Our study, conducted in the intermittent Krčić River, provides novel insights into the freshwater macroinvertebrate community patterns and processes in intermittent rivers in the Dinaric Western Balkan ecoregion (Pešić et al., 2019). The community was primarily composed of aquatic insects, and besides many IR-typical species, a large number of perennial flow indicators were recorded (Straka et al., 2019). Seasonality was inferred as the most important factor determining community composition, diversity and abundance. This was also supported by the ecological trait information, i.e. trophic and dispersal-related community structure of selected aquatic insect groups, i.e. the EPT and Chironomidae. By providing information on ecological species traits of IR inhabitants, our study helps providing the vital knowledge necessary to predict how stream communities and their populations will respond to increased drying (Robson et al., 2011).

4.1. Composition of aquatic insect community of an intermittent Mediterranean river in the Dinaric Western Balkans

Chironomids, the most widespread and numerous group of insects in freshwater ecosystems (Ferrington, 2007), were also the most abundant and diverse group of macroinvertebrates in the Krčić River. Some of chironomid taxa recorded in the current study are tolerant to intermittence and potentially common and abundant in IR (e.g. *Rheocricotopus*

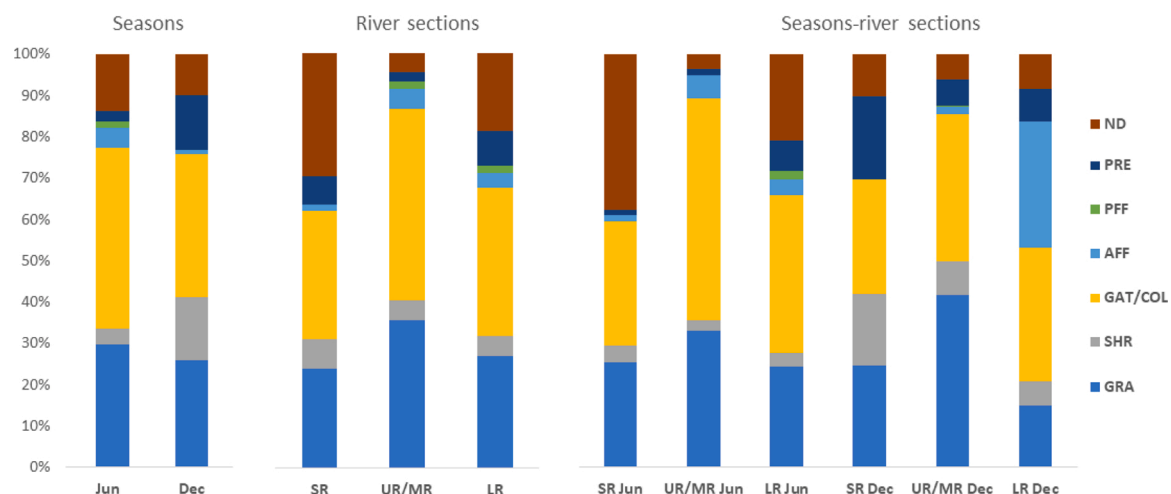


Fig. 4. Proportion of functional feeding groups of aquatic insects in two seasons at three river sections of the Krčić River (SR – spring reaches; UR/MR – upper reaches/middle reaches; LR – lower reaches; Jun – June; Dec – December; Feeding types: PRE – predators, PFF – passive filter feeders, AFF – active filter feeders, GAT/COL – gatherers/collectors, SHR – shredders, GRA – grazers, ND – no data).

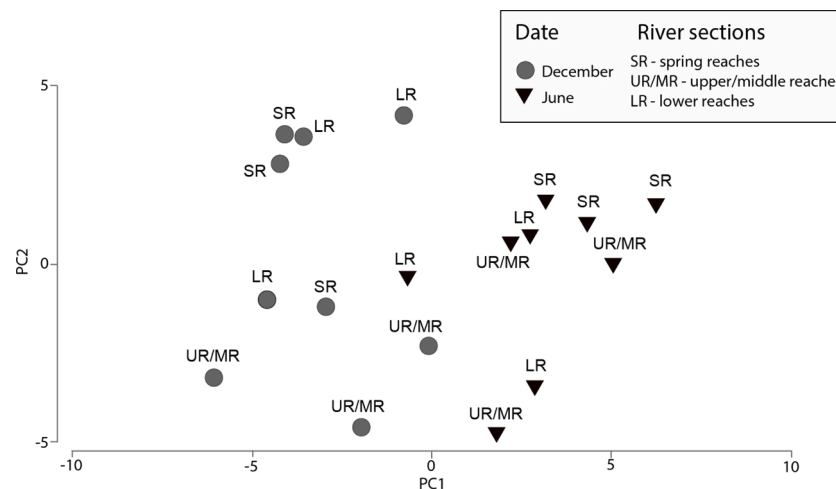


Fig. 5. Principal correspondence analysis of aquatic insect community based on proportion of nine multiple dispersal-related biological traits in different microhabitats in the Krčić River. (SR – spring reaches; UR/MR – upper reaches/middle reaches; LR – lower reaches).

effusus, *Procladius* sp., Straka et al., 2019), however, for many of them the exact process of survival in such habitats is still unknown due to the overall lack of knowledge on their taxonomy and ecology (Langton and Casas, 1998). Generally suggested adaptations to intermittency include desiccation resistant eggs (e.g. in *Micropsectra* sp., the most abundant taxon in the study; Vallenduuk and Moller Pillot, 2007) larval diapause and/or larval survival in deeper layers of sand or gravel, as well as recolonisation from nearby perennial waterbodies via passive adult dispersal (Moller Pillot, 2009). For instance, the genus *Procladius* is a taxon that cannot survive complete desiccation, hence, seasonal and spatial distribution in the Krčić River are most likely the result of recolonisation of adult females from the adjacent, perennial Krka River (Vallenduuk and Moller Pillot, 2007). The collection of the terrestrial genus *Smittia*, gives further support to its distribution in IR outside of the Central Europe (Straka et al., 2019), and represents the first record of this taxon in Croatia (Čerba et al., 2020). However, along with supporting previous findings of taxa usually recorded from temporary running waters, the current study gives important new insights into specific adaptations of chironomids to harsh environmental conditions in these habitats. For instance, *Thienemanniella vittata* (Edwards, 1924) is currently unknown from temporary streams but there are records of other *Thienemanniella* species in such habitats (Langton and Casas, 1998). However, as the genus was assessed as permanency indicator in the Central Europe (Straka et al., 2019), our results highlight the need of precise species level identification of taxa for monitoring purposes.

Among the EPT taxa, a variety of adaptations to intermittency have been observed. For instance, the mayfly community composition in temporary habitats showed to be highly related to drought advantageous species traits (discussed in detail in Řezníčková et al., 2010 and references herein). In accordance, the most abundant mayfly species in this study was the eurytopic *B. rhodani*, characterized by most of these traits (Bauernfeind and Soldán, 2012; Řezníčková et al., 2010). Moreover, Řezníčková et al. (2010) showed that *B. rhodani* and *S. aestivalis* were the most successful species for survival under the harsh conditions of the intermittent habitat. Among Plecoptera recorded here, *Isoperla tripartita* and *Perla marginata*, recorded in the Krčić river (W. Graf, personal communication), were previously reported from intermittent streams (Karaouzas et al., 2017; Řezníčková et al., 2013). On the contrary, *Perla* sp. were generally identified as inhabitants of perennial streams in the Central Europe (Straka et al., 2019), potentially indicating intraspecific ecological plasticity within taxa (e.g. Williams et al., 1995) or cryptic biodiversity typical for aquatic insects in the Balkan Peninsula (e.g. Previšić et al., 2014). *Brachyptera tristis* (W. Graf, personal communication), was previously recorded to inhabit karst spring

habitats and subterranean river bed (Sket et al., 2004). Moreover, Sket et al. (2004) reported adults to lay eggs and fulfill the life cycle in caves, indicating potential adaptations to intermittent habitats. As some *Brachyptera* species are considered as typical representatives of intermittent rivers fauna (Straka et al., 2019), thus similar preferences of their congeners are not surprising. The most abundant caddisfly species in the current study, *M. nycterobia*, is considered as intermittence indicator in the Central Europe (Straka et al., 2019), and usually occurs in temporary waterways in the Mediterranean region due to specially adapted life cycle (Graf et al., 2008; Waringer and Graf, 2011). Its resistance/resilience to drought strategy enabling large and stable populations in the adult diapause, spent mostly in nearby caves (Graf et al., 2020, 2008).

Hence, the EPT community observed in the Krčić River is mainly composed of taxa distributed in the region with adaptations enabling survival in and/or recolonisation of temporary freshwater habitats. The most surprising result, however, is the presence of the mayfly *Rh. braaschi*. Although the species' biology is still insufficiently investigated (Bauernfeind and Soldán, 2012; Vilenica et al., 2018, 2017), available studies have not reported other *Rhithrogena* species to be able to colonize intermittent streams (Buffagni et al., 2020; Řezníčková et al., 2010), but on the contrary are considered as perennial stream indicators (Straka et al., 2019). Nevertheless, certain flexibility in species' life cycle, i.e. variations in emergence period duration (Vilenica et al., 2017, 2018) could have given it the advantage in such habitats, yet, this is to be inspected in more detail. Moreover, future detail catchment inspections could answer if any perennial refuges (permanent pools or springs) could help to survive the dry phase (Love et al., 2005).

In the current study, aquatic insect taxa considered both perennial flow indicators and intermittence tolerant were recorded along the course of the Krčić River (Buffagni et al., 2020; Graf et al., 2020; Řezníčková et al., 2013; Straka et al., 2019). Higher number of both indicator taxa were recorded in June than in December, in accordance with generally higher abundance and taxa richness in June. However, in both seasons, the proportion of perennial flow indicators and intermittence tolerant taxa was approximately 2:1, suggesting the complex recolonisation patterns in the Krčić River from undiscovered perennial refugia at different spatial scales. For instance, spring-fed residual pools could present potential refugia but stay unrecorded due to limited accessibility of parts of the river course. Additionally, microbial mats, that were observed to provide shelter for the stygophilic amphipod (*Synurella ambulans*) and enable their successful retreat into underground during dry periods in the Krčić River (Žutinić et al., 2018), could also provide shelter for early instars and/or smaller insect specimens. Additionally, available information on tolerance and sensitivity to

intermittency of aquatic insects could be regionally limited (e.g. Řezníčková et al., 2013; Straka et al., 2019 have focused on temperate central European taxa), thus data on sensitivity of populations inhabiting karstic waterbodies in the Balkans merit further attention.

Considering impacts of environmental variables measured in the current study, water temperature, as one of the key factors affecting aquatic organisms (e.g. Moog and Hartmann, 2002), was shown as a good predictor of the composition of both tested communities, chironomids and the EPT according to the CCA analysis. Even though the temperature gradient recorded between sites in this study is low in both seasons, we had only point measurements, thus overall and daily variations are expectedly higher, particularly shortly before drying (in the upper reaches up to 5.5 °C and in the lower reaches up to 6.7 °C, A.B. unpublished data). Changes in the macroinvertebrate community along the course considering longitudinal zonation patterns usually reflect variability of the temperature regimes (Moog and Hartmann, 2002). However, in the current study this is observable only for a small fraction of the community (data not shown), as data on these traits are not available for Chironomidae (AQEM Consortium, 2002), the most abundant group. Additionally, conductivity was shown as important environmental factor in shaping the chironomid community, whereas pH value and distance from spring were important in shaping the EPT community. However, the EPT taxa collected in the Krčić River either have no particular pH preference (apart from avoiding low pH values), or it is not known (e.g. Buffagni et al., 2020; Graf et al., 2020). Nevertheless, distance from the spring shows a congruent pattern with the pH value, and the former was used as a proxy for the complex spatial succession of physiographic and biological patterns changing along the longitudinal river course, i.e. stream zonation patterns (Moog and Hartmann, 2002). Conductivity in rivers strongly depends on mineral composition of aquifer and its porosity (Purvanec and Andricevic, 2000). Conductivity usually exhibits slight seasonal differences, with higher values in the summer months and lower during winter, due to differences in discharge (Moss, 2010). Moreover, in karstic rivers, conductivity, pH value and CaCO₃ deposition are interdependent, and change along the course of the river (Drysdales et al., 2002). In the Krčić River, the CaCO₃ deposition evidently increases along the course, with the bedrock composed of large plates of solid tufa downstream of the spring and upper reaches (personal observation). As the composition of the substrate also plays a crucial role in determining the occurrence and distribution of aquatic organisms (Moog and Hartmann, 2002), the interconnection of few most important environmental variables is not surprising.

4.2. Seasonal and spatial dynamics of aquatic insect community in the Mediterranean intermittent river in the Western Balkans

Results of the current study show that the season is the most important factor for predicting aquatic community composition in the intermittent Krčić River most probably due to seasonality of available food resources (e.g. particularly in the lower reaches with closed-canopy riparian zone composed of popular trees) coupled with the flow intermittence (Křoupalová et al., 2011). Similarly, a previous study of IR in Montenegro demonstrated the importance of seasonality in determining the abundance of macroinvertebrate communities (Pešić et al., 2019). Such results are expected considering that life cycles of many species inhabiting IR are adapted to flow interruptions (Stubington et al., 2018). As numerous taxa emerge before the drying period and are in the egg stage during the flow re-establishment, they cannot be detected by macroinvertebrate sampling after rewetting, i.e. in autumn and winter (Bogan et al., 2015; Graf et al., 2008; Sket et al., 2004). Furthermore, the distance from the perennial rivers plays an important role in resilient taxa colonization of IR when the water habitat becomes available (Stubington et al., 2018). It is particularly important for organisms which migrate and recolonize new habitats horizontally (aerial recolonization) in adult stages, but are not good fliers, e.g. many mayflies,

stoneflies and Diptera (Stubington et al., 2018). Thus, the primarily seasonal variability is also evident when dispersal-related biological traits of dominant taxa are considered. Smaller taxa with short life cycles and poor dispersal abilities and passive dispersal strategy (i.e. Chironomidae; Armitage et al., 1995; Sarremejane et al., 2020) dominate the community in the winter, whereas higher proportion of large taxa with higher dispersal abilities and longer life cycle is observed in the summer (e.g. *M. nycterobia*). On the other hand, active dispersal mode was inferred to differ mainly in regard to river sections, with highest proportions of both aquatic and aerial active mode in the UR/MR and lowest in the SR, reflecting abundance of EPT taxa in the community at particular river reaches. Thus, observed patterns most likely reflect the combination of both, recolonisation patterns in the Krčić River and life cycles of taxa dominating the community (Sarremejane et al., 2020).

There were significant differences in the proportions of particular FFGs for all river sections, however, collectors were overall the dominant group. In previous studies, the increased number of collectors and reduced proportion of shredders indicated disturbance in the environment caused by organic pollution (Mangadze et al., 2019). However, the anthropogenic impact on the Krčić River is minimal. Nevertheless, there are examples of similar trophic structure in pristine perennial karstic streams in the region (e.g. in forest streams in the Plitvice Lakes NP, with dominance of collectors (Habdija et al., 1994). Furthermore, in the study of Álvarez and Pardo (2009), the dominance of collectors and scrapers in a temporary Mediterranean river was explained by increased primary production, compared to the input of allochthonous material. Riparian vegetation is scarce along the Krčić River (apart from the lower reaches), facilitating light penetration and algal growth, which most likely also supports higher abundance of grazers (Cañedo-Argüelles et al., 2015). Algae and algal derived detritus are easily assimilated and decomposed faster by animals (Álvarez and Pardo, 2009), which is why algae are the main food source for grazers and collectors in IR (Fisher et al., 1982). In addition to algae and allochthonous material, moss is an important source of food as well, especially for Ephemeroptera (Kalachova et al., 2011). Furthermore, metrics based on FFG were recently identified to show similar responses to anthropogenic impacts in perennial and intermittent rivers, thus providing reliable biological assessments in IR and a potential to act as universal metrics in such naturally disturbed ecosystems (Soria et al., 2020).

Results of the current study highlight the variability in macroinvertebrate community in different microhabitats in IRs, at both, spatial and temporal scales. Variability in macroinvertebrate (and aquatic insects) abundance at both scales was mostly attributable to differences between the two microhabitat pairs, i.e. moss-xyllal and moss-boulders/cobbles. By showing significant differences between macroinvertebrate communities sampled in organic-based and mineral-based substrates, our results indicate that microhabitats support specific communities determined primarily by the nature of the substrate (García-Roger et al., 2011). This is further supported by the PCA based on representation of dispersal-related traits of aquatic insect community, implying that besides season as the main grouping variable, microhabitat also seems to influence the patterns (i.e. some of samples from moss habitats cluster regardless of the season and river section). However, since microhabitat composition and proportions change along the course, the same species could select within a different microhabitat subset at different sites, thus limiting the full comparison among sites. Different microhabitats provide a variety of sites for foraging, resting, reproducing, development, as well as shelter from predators or unfavorable environmental conditions (Armitage and Cannan, 2000). Accordingly, there are numerous studies on the importance of microhabitats for community composition in perennial rivers (e.g. Costa and Melo, 2008; Lamouroux et al., 2004). However, similar research conducted in intermittent water bodies is scarce (but see Boulton and Suter, 1986; Fairchild et al., 2003; García-Roger et al., 2011) and the results are difficult to compare because of a great variety of IR ecosystems (Acuña et al., 2005).

5. Concluding remarks: implications for monitoring

Due to peculiar dynamics of IRs, it is necessary to develop an adequate management and monitoring procedure (Acuña et al., 2014). Our study highlights the great variability of IRs in the Mediterranean region of the Western Balkans, which should be considered during monitoring and management. Seasonality is the most evident characteristic of IRs in this region, so ideally aquatic community should be sampled before and after the dry period for monitoring purposes. If, however, out of practical reasons, only one sample per year (or few years) is possible, different seasons should not be directly compared. Additionally, development of the IR-specific metrics, for instance, using the dominant but insufficiently investigated groups such as the Chironomidae (e.g. Dorić et al., 2021) should be encouraged. Such IR-tailored monitoring procedures would allow timely response and prevention or mitigation of adverse effects on these ecosystems.

CRedit authorship contribution statement

Ivana Grgić: Data curation, Investigation, Methodology, Visualization, Software, Writing - original draft, Writing - review & editing. **Marina Vilenica:** Data curation, Investigation, Methodology, Writing - original draft. **Andreja Brigić:** Conceptualization, Investigation, Software, Writing - original draft. **Valentina Dorić:** Data curation, Investigation, Methodology, Writing - original draft. **Zlatko Mihaljević:** Conceptualization, Investigation, Writing - original draft. **Ana Previšić:** Conceptualization, Investigation, Methodology, Visualization, Software, Writing - original draft, Writing - review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found, in the online version, at doi:<https://doi.org/10.1016/j.limno.2022.125953>.

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